MTH107 2024S1 Exam

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*	MTH316	Multivariable Calculus
*	MTH321	Engineering Mathematics 2
*	MTH355	Basic Mathematical Optimisation
*	MTH357	Applied Regression Analysis I

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SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

MTH107

Examination - January Semester 2024

Calculus I

Tuesday, 16 April 2024 **01:00 pm - 03:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FIVE (5)** questions and comprises **FOUR (4)** printed pages (including cover page).

2. You must answer **ALL** questions.

3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.

4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1a

Determine

$$\lim_{x \to 1} \frac{x^{2024} - 1}{x^{1012} - 1}$$

<u>without using L'Hopital's Rule</u>. Show your steps clearly.

(10 marks)

Question 1b

Show that there exists a negative real root for the equation $\sin x = x^2 - \pi$. Justify your answer.

(10 marks)

Question 2a

Define the function $f : \mathbb{R} \longrightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} x^3 \cos(\frac{\pi}{x^4}) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

Discuss whether f is continuous at $x = 0$.

(10 marks)

Question 2b

Define the function $g : \mathbb{R} \longrightarrow \mathbb{R}$ by

$$g(x) = \begin{cases} ax + b, & \text{when } x \leq 1 \\ ax^2 + x + 2b, & \text{when } x > 1. \end{cases}$$

Determine all value(s) of $a, b \in \mathbb{R}$ such that the function g is differentiable for all $x \in \mathbb{R}$.

(10 marks)

Question 3a

Determine

$$\lim_{x \to 2024} \frac{x - 2024}{\ln x - \ln 2024}.$$

Show your steps clearly.

(8 marks)

Question 3b

Use the Mean Value Theorem to show that

$$8 < \sqrt[3]{513} < 8 + \frac{1}{192}.$$

(12 marks)

Question 4a

Define the function $h : \mathbb{R} \longrightarrow \mathbb{R}$ by

$$h(x) = e^{x^2}.$$

Determine the value of $h^{(2024)}(0)$, expressing your answer in the form $\dfrac{a!}{b!}$.

(6 marks)

Question 4b

Define the function $f : \mathbb{R} \longrightarrow \mathbb{R}$ by

$$f(x) = \frac{\cos x}{e^x}.$$

Determine the Taylor expansion of the function f about $x = 0$, up to and including the term in x^2. Use this Taylor expansion to estimate the value of

$$\frac{\cos 0.1}{e^{0.1}},$$

and estimate the error in your evaluation.

(14 marks)

Question 5

Define the function $f : \mathbb{R} \longrightarrow \mathbb{R}$ by

$$f(x) = \frac{2024 + x}{x^2}.$$

Question 5a

Apply the first derivative test to compute the intervals on which f is decreasing.

(10 marks)

Question 5b

Determine the critical point(s) of f. Apply relevant derivative tests and discuss the nature of the critical point(s). Show your steps clearly.

(5 marks)

Question 5c

Determine the horizontal and vertical asymptotes of f if they exist. Justify your answers.

(5 marks)

----- END OF PAPER -----

1(a) $\lim_{x \to 1} \frac{x^{2024} - 1}{x^{1012} - 1} = \lim_{x \to 1} \frac{(x^{1012} - 1)(x^{1012} + 1)}{x^{1012} - 1}$

$= \lim_{x \to 1} (x^{1012} + 1)$

$= 2$

(b) $\sin x = x^2 - \pi$

$\sin x - x^2 + \pi = 0$

let $f(x) = \sin x - x^2 + \pi$

$f(0) = \sin 0 - 0 + \pi = \pi$

$f(-\pi) = \sin(-\pi) - (-\pi)^2 + \pi$

$= 0 - \pi^2 + \pi$

$= \pi - \pi^2$

since $f(0) > 0$ and $f(-\pi) < 0$

By Intermediate Value Theorem, $\exists c$ on $(-\pi, 0)$ such that $f(c) = 0$

$\therefore$ There exists a negative real root for $\sin x = x^2 - \pi$

2(a)

$$-|x^3| \le -|x^3 \cos\frac{\pi}{x^4}| \le x^3 \cos\frac{\pi}{x^4} \le |x^3 \cos\frac{\pi}{x^4}| \le |x^3|$$

$$\Rightarrow -|x^3| \le x^3 \cos\frac{\pi}{x^4} \le |x^3|$$

$$\lim_{x\to 0} -|x^3| \le \lim_{x\to 0} x^3 \cos\frac{\pi}{x^4} \le \lim_{x\to 0} |x^3|$$

$$0 \le \lim_{x\to 0} x^3 \cos\frac{\pi}{x^4} \le 0$$

By Squeeze Theorem, $\lim_{x\to 0} x^3 \cos\frac{\pi}{x^4} = 0$

$$\lim_{x\to 0} f(x) = \lim_{x\to 0} x^3 \cos\frac{\pi}{x^4} = 0 = f(0)$$

Hence, $f(x)$ is continuous at $x=0$

(b) $$g(x) = \begin{cases} ax+b & \text{if } x \ge 0 \\ ax^2+x+2b & \text{if } x<0 \end{cases}$$

For $g(x)$ to be differentiable, $g(x)$ has to be continuous

$$g(0) = a(0)+b = b$$

$$\lim_{x\to 0^-} g(x) = \lim_{x\to 0^-} (ax^2+x+2b) = 2b$$

let $\lim_{x\to 0^-} g(x) = g(0)$

$$2b = b$$

$$\therefore b = 0$$

For $g(x)$ to be differentiable at $x=0$

let $$\lim_{h\to 0^+} \frac{g(h)-g(0)}{h-0} = \lim_{h\to 0^-} \frac{g(h)-g(0)}{h-0}$$

$$\lim_{h\to 0^+} \frac{g(h)-g(0)}{h-0} = \lim_{h\to 0^+} \frac{ah}{h} = a$$

$$\lim_{h\to 0^-} \frac{g(h)-g(0)}{h-0} = \lim_{h\to 0^-} \frac{ah^2+h}{h} = \lim_{h\to 0^-} (ah+1) = 1$$

$$\therefore a = 1$$

$\therefore$ $g(x)$ is differentiable for all $x \in \mathbb{R}$ when $a=1, b=0$

$$\therefore g(x) = \begin{cases} x & \text{if } x \ge 0 \\ x^2+x & \text{if } x<0 \end{cases}$$

3(a) $\lim_{x \to 2024} \frac{x-2024}{\ln x - \ln 2024} = \lim_{x \to 2024} \frac{1}{(1/x)}$ $\frac{0}{0}$ form, apply L'Hopital Rule

$= \lim_{x \to 2024} x$

$= 2024$

(b) Show $8 < \sqrt[3]{513} < 8 + \frac{1}{192}$ using MVT

let $f(x) = x^4$, $f'(x) = 4x^3$

By MVT, $\exists c$ on $(8, 8+\frac{1}{192})$ such that

$$f'(c) = \frac{f(8+\frac{1}{192}) - f(8)}{(8+\frac{1}{192}) - 8}$$

$$\begin{aligned} 4c^3 &= 192\left((8+\tfrac{1}{192})^4 - 8^4\right) \\ &= 192\left[4(8)^3(\tfrac{1}{192}) + 6(8)^2(\tfrac{1}{192})^2 + 4(8)(\tfrac{1}{192})^3 + (\tfrac{1}{192})^4\right] \\ &= 2048 + 2 + \frac{32}{192^2} + \frac{1}{192^3} \\ &= 2050 + \frac{32}{192^2} + \frac{1}{192^3} \end{aligned}$$

$\Rightarrow 4c^3 < 2052$

$c^3 < 513$

$8 < c < \sqrt[3]{513}$

$\therefore 8 < \sqrt[3]{513}$

~~Hence, $8 < \sqrt[3]{513} < 8 + \frac{1}{192}$~~

$$\begin{aligned} (8+\tfrac{1}{192})^3 &= \binom{3}{0}(\tfrac{1}{192})^0(8)^3 + \binom{3}{1}(\tfrac{1}{192})^1(8)^2 + \binom{3}{2}(\tfrac{1}{192})^2(8)^1 + \binom{3}{3}(\tfrac{1}{192})^3(8)^0 \\ &= 8^3 + \frac{192}{192} + \frac{24}{192^2} + \frac{1}{192^3} \\ &= 513 + \frac{24}{192^2} + \frac{1}{192^3} \end{aligned}$$

$\Rightarrow (8+\frac{1}{192})^3 > 513$

$\therefore 8 + \frac{1}{192} > \sqrt[3]{513}$

Hence $8 < \sqrt[3]{513} < 8 + \frac{1}{192}$

4(a) $h(x) = e^{x^2}$

By Power Series Expansion,

$$e^x = \sum_{n=1}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots$$

$$e^{x^2} = \sum_{n=1}^{\infty} \frac{(x^2)^n}{n!} = \sum_{n=1}^{\infty} \frac{x^{2n}}{n!} = 1 + \frac{x^2}{1!} + \frac{x^4}{2!} + \dots + \frac{x^{2024}}{1012!} + \dots$$

By Taylor Expansion,

$$h(x) = \sum_{n=0}^{\infty} \frac{h^{(n)}(0)}{n!} x^n$$

$$= h(0) + \dots + \frac{h^{(2024)}(0)}{2024!} x^{2024} + \dots$$

Comparing coefficients of the x^{2024} term,

$$\frac{h^{(2024)}(0)}{2024!} = \frac{1}{1012!}$$

$$\therefore h^{(2024)}(0) = \frac{2024!}{1012!}$$

(b) $f(x) = e^{-x}\cos(x)$

$$f'(x) = -e^{-x}\cos(x) - e^{-x}\sin(x)$$

$$= -e^{-x}(\sin x + \cos x)$$

$$f''(x) = e^{-x}(\sin x + \cos x) - e^{-x}(\cos x - \sin x) = 2e^{-x}\sin x$$

$f(0) = 1 \quad f'(0) = -1 \quad f''(0) = 0$

By Taylor Expansion around $x=0$

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \dots = 1 - x + \dots$$

$$\Rightarrow T_2(x) = 1 - x$$

$$\frac{\cos 0.1}{e^{0.1}} = f(0.1) \approx T_2(0.1) = 1 - 0.1 = 0.9$$

$$f^{(3)}(x) = 2e^{-x}\cos x - 2e^{-x}\sin x$$

error for $\frac{\cos 0.1}{e^{0.1}} = |R_2(0.1)|$

$= \left|\frac{f^{(3)}(c)}{3!}(0.1)^3\right|$ for some $0 < c < 0.1$

$= \left|\frac{2e^{-c}\cos c - 2e^{-c}\sin c}{3!}(0.1)^3\right|$

$= \frac{0.001}{3} e^{-c}(\cos c - \sin c)$ because $\cos c > \sin c$ for $0 < c < 0.1$

$e^{-c}(\cos c - \sin c)$ is max at $c = 0$ on $[0, 0.1]$

max error estimate $= \frac{0.001}{3} e^{0}(\cos 0 - \sin 0)$

$= \frac{0.001}{3}$

5(a) $f(x) = \frac{2024+x}{x^2} = \frac{2024}{x^2} + \frac{1}{x}$

$f'(x) = -2(2024)x^{-3} + (-1)x^{-2}$

$= -\frac{4048}{x^3} - \frac{1}{x^2}$

let $f'(x) > 0$

$-\frac{4048}{x^3} - \frac{1}{x^2} > 0$

$\frac{4048}{x^3} + \frac{1}{x^2} > 0$

case $x > 0$:

$x^3 > 0$

$x^3\left(\frac{4048}{x^3} + \frac{1}{x^2}\right) > 0$

$4048 + x > 0$

$x > -4048$

case $x < 0$

$x^3 < 0$

$x^3\left(\frac{4048}{x^3} + \frac{1}{x^2}\right) < 0$

$4048 + x < 0$

$x < -4048$.

$\therefore f'(x) > 0$ when $x > 0$ or $x < -4048$

interval where $f(x)$ is increasing $= \{x : x > 0 \text{ or } x < -4048\}$

(b) $f^{(2)}(x) = -4048(-3)x^{-4} - (-2)x^{-3} = \frac{2}{x^3} + \frac{12144}{x^4}$

let $f'(x) = 0$

$-\frac{4048}{x^3} - \frac{1}{x^2} = 0$

$4048 + x = 0$

$x = -4048$

$f(-4048) = -\frac{1}{8096}$

$f''(-4048) = 1.5075 \times 10^{-11} > 0$

$\therefore$ the critical point $(-4048, -\frac{1}{8096})$ is a minima

(c) $f(x)$ is a rational function $\Rightarrow$ $x = 0$ is vertical asymptote

$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \frac{2024+x}{x^2}$

$= \lim_{x\to\infty} \frac{1}{2x}$

$= 0$

$\frac{\infty}{\infty}$ form, apply L'Hopital Rule

$\lim_{x\to-\infty} f(x) = \lim_{x\to-\infty} \frac{2024+x}{x^2}$

$= \lim_{x\to-\infty} \frac{1}{2x}$

$= 0$

$-\frac{\infty}{\infty}$ form, apply L'Hopital Rule

Horizontal asymptote is $y = 0$